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CAPACITIVE FLUID DETECTION SYSTEM FOR A PUMP

Abstract

Disclosed herein are a pump and a method for controlling the pump for detecting a fluid around the pump based on a value detected by a capacitive sensor. The pump may include a motor housing accommodating a motor configured to drive an impeller to discharge fluid. The pump may include a pump cap, located above the motor housing, accommodating a controller configured to drive the motor. The pump may include a sensor box, provided on a side of the pump, accommodating a capacitive sensor. The controller may be configured to drive the motor after determining that a value detected by the capacitive sensor indicates that the fluid is present around the pump.

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Background/Summary

BACKGROUND

[0001] A pump is a device that uses mechanical, hydraulic, pneumatic, or electrical energy to move fluids from one place to another, often against a pressure difference or a gravitational force. Pumps are used to provide adequate drainage for both commercial and residential uses. Examples of different types of pumps include sump, sewage, and drainage pumps.

[0002] A sump pump is a pump installed in a basin or pit below the ground level, usually in basements, crawl spaces, or other low-lying areas, to collect and remove excess water that may accumulate from rain, groundwater, or flooding. Sump pumps can be placed inside or above a sump basin. Sump pumps prevent water damage, mold growth, and structural problems in buildings and foundations.

[0003] A sewage pump is a pump typically used to transport wastewater or sewage from a building to a septic tank, a sewer system, or a treatment plant. Sewage pumps can handle solid waste, organic matter, and other contaminants that may clog or damage other types of pumps. Some sewage pumps are equipped with a grinder, that is, a cutting mechanism to reduce the size of the solids suspended in the fluid. Sewage pumps prevent sanitary and health hazards, odors, and backups in plumbing systems.

[0004] A drainage pump is a pump used to remove water from surfaces or subsurfaces, such as fields, gardens, roads, roofs, or mines. Drainage pumps can be portable (moved from one location to another) or fixed (installed permanently in a specific location). Drainage pumps prevent flooding, erosion, and waterlogging in various settings.

[0005] Regardless of the environment in which the pumps are used, the important thing is to drain at the right time. For example, if the pump does not drain even though there is fluid in the environment, it will not fulfill its intended purpose. On the other hand, if the pump attempts to drain even though there is no fluid in the surrounding area, it will unnecessarily shorten the life of the product.

[0006] There is a known method of controlling the pump with floats in order to drain fluid at the appropriate time. In this method, a float is placed in the basin so that the float rises as fluid accumulates around the pump. When the float rises, a mechanical switch is turned on to activate the pump, thus enabling the pump to be controlled according to the amount of fluid around the pump.

[0007] However, control using floats tends to require a large system configuration. In addition, the control using floats has many moving parts, which may not be easy to maintain. Thus, improved systems and methods for controlling pumps are needed.

SUMMARY

[0008] According to an embodiment, a pump is provided. The pump may include a motor housing accommodating a motor configured to drive an impeller to discharge fluid. The pump may include a pump cap, located above the motor housing, accommodating a controller configured to drive the motor. The pump may include a sensor box, provided on a side of the pump, accommodating a capacitive sensor. The controller may be configured to drive the motor after determining that a value detected by the capacitive sensor indicates that the fluid is present around the pump.

[0009] System, device, and computer program product aspects are also disclosed.

[0010] Further features and advantages, as well as the structure and operation of various aspects, are described in detail below with reference to the accompanying drawings. It is noted that the specific aspects described herein are not intended to be limiting. Such aspects are presented herein for illustrative purposes only. Additional aspects will be apparent to persons skilled in the relevant art(s) based on the teachings contained herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The accompanying drawings are incorporated herein and form a part of the specification.

[0012] FIG. 1 is a perspective view of a pump, according to some embodiments.

[0013] FIG. 2 is a cross section of the pump, according to some embodiments.

[0014] FIG. 3 is an enlarged diagram illustrating an internal structure of a pump cap, according to some embodiments.

[0015] FIG. 4 is an enlarged diagram illustrating an internal structure of a sensor box, according to some embodiments.

[0016] FIG. 5 is a flowchart for a method for operating the pump, according to some embodiments.

[0017] In the drawings, like reference numbers generally indicate identical or similar elements.

Additionally, generally, the left-most digit(s) of a reference number identifies the drawing in which the reference number first appears.

DETAILED DESCRIPTION

[0018] This disclosure generally relates to fluid detection around a pump. In particular, this disclosure may relate to technology that makes it possible to control the pump to run or stop in response to changes in a surrounding environment by installing a capacitive sensor on the pump. A capacitive sensor may reduce the need for mechanical parts, which may improve reliability of the pump.

[0019] Another aspect of this disclosure may relate to a preventive function of the pump. If an environment in which the pump is placed is poor or if maintenance is inadequate, the pump may tip over or gunk may adhere to the pump, preventing it from performing as intended. To address these problems, according to this disclosure, the pump can perform the preventive function by detecting tipping over and gunk buildup based on the value detected by the capacitive sensor.

[0020] FIG. 1 is a perspective view of a pump **100**, according to some embodiments. The pump **100** may include a pump cap **110**, a power cord **120**, a sensor box **130**, a motor housing **150**, a discharge outlet **170**, a pump housing **190**, and a base **195**.

[0021] The pump cap **110** may be located above the motor housing **150**. The pump cap **110** may accommodate electrical components. The pump cap **110** may be made of non-conductive material, such as plastics or polymers. Details of the pump cap **110** are described below.

[0022] The power cord **120** may penetrate the pump cap **110** and supply power to the pump **100**. The top of the power cord **120** may be connected to an external power source. The external power source may, for example, be household electrical power, e.g., 120 volts alternating current.

[0023] The sensor box **130** may be provided on a side of the pump cap **110**. The sensor box **130** may accommodate a capacitive sensor. The capacitive sensor may be a non-contact sensor that can detect surrounding objects by detecting changes in ambient capacitance. From the pattern of capacitance changes, it is possible to estimate what kind of objects are in the surrounding area.

[0024] The sensor box **130** may be made of non-conductive material, such as plastics or polymers. Screws **135** secure a lid **133** of the sensor box **130**, and a screw **137** secures the sensor box **130** to the pump cap **110**. Both screws **135** and **137** may be made of metal. Details of the sensor box **130** are described below.

[0025] The motor housing **150** may accommodate a motor to drive the pump. Details of the motor housing **150** are described below.

[0026] The discharge outlet **170** may discharge fluid pumped by the pump **100**. A discharge pipe, not shown in FIG. 1, may be connected to the discharge outlet **170** to discharge the fluid to the outside.

[0027] The pump housing **190** may be provided underside of the motor housing **150**. The pump housing **190** may accommodate an impeller driven by the motor. The base **195** may be provided underside of the pump housing **190**.

[0028] FIG. 2 is a cross section of the pump **100** along the 2-2 cross-section of FIG. 1, according to

some embodiments. The pump cap **110** may accommodate a run capacitor **212** and a printed circuit board (PCB) **214**.

[0029] The run capacitor **212** may provide starting torque to the motor by supplying voltage to the motor.

[0030] The PCB **214** may be a board where a hall sensor, a timer, a controller, and other electronic components necessary to control pump **100** are mounted. The hall sensor may detect a rotational speed of the motor, as a speed sensor. The timer may measure the time required to control the pump **100**. The controller may control the electronic components that make up the pump **100**.

[0031] The motor housing **150** may accommodate a motor **250**. The motor **250** may include a magnet **252**, a rotor **254**, and a stator **256**. The magnet **252** is located at the bottom of the PCB **214**. The hall sensor located on the PCB **214** detects the magnetic force of the magnet **252** to sense the rotational speed of the motor **250**. The rotor **254** may rotate when power is supplied to the motor **250**. The stator **256** may accommodate a laminated core and coils and generate a magnetic field to rotate the rotor **254**. A portion of the motor housing **150** may be filled with motor oil to lubricate the components of the motor **250**.

[0032] The pump housing **190** may accommodate an impeller **292**. The impeller **292** may be connected to the rotor **254** and rotate as the rotor **254** rotates. As the impeller **292** rotates, the fluid around pump **100** may be discharged through the discharge outlet **170**.

[0033] FIG. **3** is an enlarged diagram illustrating an internal structure of the pump cap **110**, according to some embodiments. As illustrated in FIG. **3**, there may be a hole **316** between the pump cap **110** and the motor housing **150**. The motor oil may enter the pump cap **110** through the hole **316** when the pump **100** is tipped over.

[0034] The sensor box **130** may include a seal **332** between the sensor box **130** and the pump cap **110**. The seal **332** may prevent the motor oil from entering the sensor box **130**. The seal **332** may be made of silicone.

[0035] The sensor box **130** may also include a gasket **334** between the lid **133** and the sensor box **130**. The gasket may prevent the fluid from entering the sensor box **130**. The gasket **334** may be made of rubber.

[0036] FIG. **4** is an enlarged diagram illustrating an internal structure of the sensor box **130**, according to some embodiments. A PCB support part **432** may be an about inverted L-shaped bracket. One side of the PCB support part **432** may be secured to the sensor box **130** by a screw **433**, approximately parallel to the installation surface of the pump **100**. A sensor **434** may be secured to the other side of the PCB support part **432** by screws **438**. The sensor **434** may be a capacitive sensor. As represented in FIG. **4**, the one side and the other side of the PCB support part **432** are nearly perpendicular to each other. The PCB support part **432** may be made of non-conductive material, such as plastics or polymers. A rubber gasket **440** may securely attach the PCB support part **432** and a sensor **434** to the sensor box **130**.

[0037] A sensing surface **436** may be secured by the PCB support part **432** to contact an inner side of the sensor box **130** so that the sensing surface **436** faces outward of the pump **100**. In this way, the sensor **434** can detect changes in the external environment of the pump **100**. The elasticity of the rubber gasket **440** may allow the sensing surface **436** to more firmly contact the inner surface of the sensor box **130**.

[0038] As shown in FIG. **4**, the sensor box **130** or the sensing surface **436** may be located on the side of the pump cap **110**. In this way, it is possible to detect an abnormal situation where the fluid is coming close to the pump cap **110**, since the sensing surface **436** can detect changes in capacitance around the pump cap **110**. Also, it is possible to detect an abnormal situation where the pump **100** has tipped over. This is because if the pump **100** has tipped over, the motor oil may enter the pump cap **110** and the sensing surface **436** may detect changes in capacitance inside the pump cap **110**. Further, it is possible to prevent gunk from sticking to the sensor box **130**, since the fluid is unlikely to reach the sensor box **130** during normal operation.

[0039] As shown in FIG. 4, the sensor box **130** or the sensing surface **436** may be located above the motor housing **150**. In this way, a negative influence from the motor housing **150** to the sensor **434** can be reduced. As described, even if the lid **133** is installed above the pump cap **110**, the PCB support part **432** allows the sensing surface **436** to be installed side of the pump cap **110**.

[0040] As shown in FIG. 4, the lid **133** may be placed above the pump cap **110**. In this way, the sensor box **130** can be accessed even if the fluid has reached a height above the pump cap **110**.

[0041] FIG. 5 is a flowchart for a method **500** for operating the pump **100**, according to some embodiments. Method **500** can be performed by processing logic that can comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions executing on a processing device), or a combination thereof. It is to be appreciated that not all steps may be needed to perform the disclosure provided herein. Further, some of the steps may be performed simultaneously, or in a different order than shown in FIG. 5 as will be understood by a person of ordinary skill in the art. Method **500** shall be described with reference to FIGS. 1-4.

However, method **500** is not limited to that example embodiment. Also, although descriptions of the method **500** often uses the pump **100** as the operating entity, it will be clear to those skilled in the art that the actual operating entity can be a controller or the sensor **434** or other component of the pump **100**.

[0042] At **502**, the pump **100** may be powered on. The pump **100** may be powered on by the power cord **120** connected to a power source.

[0043] At **504**, the pump **100** may set a startup delay. The startup delay may be a period of time during which pump **100** forces the motor **250** to rotate after powered on.

[0044] At **506**, the pump **100** may turn the motor **250** on. The motor **250** may run during the startup delay period to drain the fluid that has accumulated before the power is turned on and also to detect abnormalities during initial operation.

[0045] At **508**, the pump **100** may determine whether a rotational speed exceeds a predetermined speed. For example, the controller can 1) detect ten rising edges of the hall sensor provided on the PCB **214** to determine the rotational speed of the motor **250** 2) detect ten rising edges of an AC line connected to the motor **250** to determine the predetermined speed. After detecting both series of edges, the controller may calculate the cycle of each edge and determine “Yes” in **508** if the cycle of the hall sensor edge minus the cycle of the AC line edge is less than one-third the cycle of the AC line edge.

[0046] The above calculation method is only an example. The operation in **508** may be explained that the pump **100** detects the presence or absence of the fluid around the pump **100** based on the rotational speed of the motor **250**. This is because the relationship between the frequency given to the motor **250** and the frequency at which the motor **250** is actually rotating depends on how much fluid the impeller **292** is stirring. In other words, the decision of **508** can be determined in other ways, e.g., based on the absolute speed of the impeller **292** or the resistance the impeller **292** receives from the fluid.

[0047] During **506** and **508**, the pump **100** may detect abnormal operation. For example, if one of the periods of the AC line and the hall sensor exceeds 15 Hz, the pump **100** may determine that there is a fault in signal detection. This is because it may indicate that either the current to the motor **250** or the rotation of the impeller **292** is significantly slower. In such a case, pump **100** may immediately stop the motor **250**, set a new startup delay, and the operation returns to **506** after a predetermined delay (e.g. sixty seconds) has been elapsed.

[0048] At **510**, if the pump **100** determined that the rotational speed does not exceed the predetermined speed at **508**, the pump **100** may determine whether the predetermined duration has been elapsed since the last startup delay. The predetermined duration may be fifteen seconds. If the pump **100** determines that the predetermined duration has not been elapsed, the operation returns to **506**.

[0049] During **508** and **510**, the pump **100** may detect abnormal operation. For example, if the

cycle of the hall sensor is greater than or equal to twice the cycle of the AC line, the pump **100** may determine that there is a slippage of the impeller **292**. In such a case, pump **100** may immediately stop the motor **250**, set a new startup delay, and the operation returns to **506** after a predetermined delay (e.g. ninety seconds) has been elapsed.

[0050] At **512**, if the pump **100** determines that the predetermined duration has been elapsed at **510**, the pump may clear the startup delay and the operation returns to **506**.

[0051] At **514**, if the pump **100** determined that the rotational speed exceeds the predetermined speed at **508**, the pump **100** may determine whether the startup delay has been cleared. If the startup delay has not been cleared, the operation proceeds to **510**.

[0052] At **516**, if the pump **100** determined that the startup delay has been cleared, the pump may turn the motor **250** off and set a new startup delay. In this way, if the rotational speed of the motor **250** is greater than a predetermined speed, the motor **250** stops rotating because the fluid stirred by the impeller **292** has most likely decreased. The pump **100** may set a five seconds delay between turning off the motor **250** and setting the new startup delay.

[0053] At **518**, the pump **100** may startup a three-minute timer. The three-minute period may be intended to prevent the motor **250** from turning on and off excessively in short periods of time. Therefore, the duration of the timer started in **518** is not limited to three minutes, but can be five minutes, eight minutes, or any other arbitrary duration.

[0054] At **520**, the pump **100** may detect an environment capacitance. The environment capacitance may be a capacitance value detected by the sensor **434**. The sensing surface **436** of the sensor **434** can sense the capacitance value of the environment surrounding the pump **100**, as described using FIGS. 1-4. How the capacitance value detected by the sensor **434** is quantified depends on the type of sensor **434**. As an example, this disclosure describes the use of the sensor **434** that detects a capacitance value of approximately **100** or more when the majority of the area around the pump **100** is occupied by air.

[0055] At **522**, the pump **100** may record the capacitance value detected in the previous operation as a first value. The first value recorded in this way may serve as a reference value detected in a condition where there is a high probability that no fluid is present in the surrounding area, since the rotational speed exceeds the predetermined speed. The pump **100** then may determine whether or not fluid is present in the surrounding area by comparing it to the value detected after the first value as described below.

[0056] At **524**, the pump **100** may detect an environment capacitance and record the detected capacitance value as a second value. As explained, the first value may be the value detected in the condition where there is a high probability that no fluid is present in the surrounding area. However, over time, fluid may accumulate around the pump **100**. Therefore, the pump **100** may detect the environment capacitance as the second value after detecting the first value, allowing fluid detection based on changes in the capacitance value.

[0057] At **526**, the pump **100** may determine whether the second value is less than or equal to 0.9 times the first value over 3 seconds. If the second value is less than or equal to 0.9 times the first value over 3 seconds, the operation returns to **506** to drive the motor **250**. The decrease in the second value detected after the first value may mean that the fluid is present around the pump **100** over time. Therefore, the pump **100** may drive the motor **250** to discharge the fluid. The coefficient value of 0.9 is an example and may be changed depending on the application and location of use of the pump **100**. The determination of **526** may be made simply based on the difference between the first and second values without using coefficients.

[0058] At **528**, if the pump **100** determined “no” at **526**, the pump **100** may determine whether the three-minute timer has been elapsed. This operation may prevent an operation in **536** from being processed an excessive number of times in a short period of time and prevent the motor **250** from being turned on and off frequently.

[0059] At **530**, if the pump **100** determined the three-minute timer has not been elapsed, the pump

100 may set a five seconds delay. In this way, the pump **100** can prevent excessive acquisition of new environment capacitance.

[0060] At **532**, the pump **100** may detect an environment capacitance and record the detected capacitance value as a third value. In this way, the first value, which is the reference value, can be updated at **534** described below.

[0061] At **534**, the pump **100** may determine whether the third value is greater than or equal to the first value. The fact that the third value is greater than or equal to the first value may indicate that the fluid is less likely to be present around the pump **100**.

[0062] If the third value is greater than or equal to the first value, the operation returns to **522** and the pump **100** may overwrite the first value using the third value. In this way, the first value, which may be a reference value, can be updated to a higher value in response to changes in the environment, so that the motor **250** is rotated more actively. It also prevents the motor **250** from not rotating if, for some reason, the first value is detected as smaller than it actually is. If the third value is less the first value, the operation returns to **524**.

[0063] At **536**, if the pump **100** determined “yes” at **528**, the pump **100** may determine whether the second value is greater than or equal to a predetermined threshold. In this disclosure, **90** may be used as the predetermined threshold as one example. The predetermined threshold in **536** may be above the value that the sensor **434** detects when there is some foreign substance (e.g., debris, gunk, motor oil, floor surface), around the sensor box **130**, but below the value it detects when most of the area around the pump **100** is atmosphere. The value that the sensor **434** detects when there is some foreign substance may be higher than the value that the sensor **434** detects when the fluid is present around the pump **100**.

[0064] The operation **536** may be called a protective function. The protective function can protect the pump **100** from accidents. For example, by operation **536**, the pump **100** can drive the motor **250** after detecting that the pump **100** is tipped over. As explained above, at least a portion of the motor housing **150** is filled with the motor oil and the motor oil may enter the pump cap **110** as the pump **100** tips over. Then the pump **100** can detect that the value detected by the sensor **434** indicates that the motor oil is present around the sensor box **130** (e.g., the value ≤ 90). In other words, it can be inferred that the pump **100** has tipped over based on the value detected by the sensor **434**.

[0065] The protective function can also detect the presence of the foreign substance around the sensor box **130**. If debris or gunk adheres to the sensor box **130** due to long-term use of the pump **100**, the reliability of the values detected by the sensor **434** may decrease. To prepare for such an eventuality, the pump **100** can determine whether the value detected by the sensor **434** indicates that a foreign substance has been detected (e.g., the value ≤ 90).

[0066] If the pump **100** determines that the second value is greater than or equal to the predetermined value, in other words determines that the protective function may be required, the operation may return to **506** and the pump **100** may drive the motor **250**. If the operation transitions from **536** to **506**, it is likely that the protective function is required, so a special action may be taken by the pump **100** at **506**. For example, the pump **100** may sound an alarm or perform other actions to inform the user of the pump **100** of an abnormality.

[0067] If the pump **100** determines that the second value is less than the predetermined value the operation proceeds to **530**.

[0068] It is to be appreciated that the Detailed Description section, and not any other section, is intended to be used to interpret the claims. Other sections can set forth one or more but not all exemplary embodiments as contemplated by the inventor(s), and thus, are not intended to limit this disclosure or the appended claims in any way.

[0069] While this disclosure describes exemplary embodiments for exemplary fields and applications, it should be understood that the disclosure is not limited thereto. Other embodiments and modifications thereto are possible, and are within the scope and spirit of this disclosure. For

example, and without limiting the generality of this paragraph, embodiments are not limited to the software, hardware, firmware, and/or entities illustrated in the figures and/or described herein. Further, embodiments (whether or not explicitly described herein) have significant utility to fields and applications beyond the examples described herein.

[0070] Embodiments have been described herein with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined as long as the specified functions and relationships (or equivalents thereof) are appropriately performed. Also, alternative embodiments can perform functional blocks, steps, operations, methods, etc. using orderings different than those described herein.

[0071] References herein to “one embodiment,” “an embodiment,” “an example embodiment,” or similar phrases, indicate that the embodiment described can include a particular feature, structure, or characteristic, but every embodiment can not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it would be within the knowledge of persons skilled in the relevant art(s) to incorporate such feature, structure, or characteristic into other embodiments whether or not explicitly mentioned or described herein. Additionally, some embodiments can be described using the expression “coupled” and “connected” along with their derivatives. These terms are not necessarily intended as synonyms for each other. For example, some embodiments can be described using the terms “connected” and/or “coupled” to indicate that two or more elements are in direct physical or electrical contact with each other. The term “coupled,” however, can also mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other.

[0072] The breadth and scope of this disclosure should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1.-20. (canceled)

21. A pump comprising: a motor housing accommodating a motor configured to drive an impeller to discharge fluid; a pump cap, located above the motor housing, accommodating a controller configured to drive the motor; and a sensor box, provided on a side of the pump, accommodating a capacitive sensor; wherein the controller is configured to (i) drive the motor after determining that a value detected by the capacitive sensor indicates that fluid is present around the pump, wherein determining that the value detected by the capacitive sensor indicates that the fluid is present around the pump is determined based on a difference between a first value and a second value, wherein the first value is detected by the capacitive sensor, and the second value is detected by the capacitive sensor after detecting the first value, and (ii) overwrite the first value with a third value after detecting that the third value indicates that the fluid is less likely to be present around the pump than the first value does, and the third value is detected by the capacitive sensor after detecting the second value.

22. The pump of claim 21, wherein the difference between the first value and the second value comprises a difference between the first value multiplied by a predetermined coefficient and the second value.

23. The pump of claim 21, further comprising: a speed sensor configured to detect a rotational speed of the motor; wherein the first value is detected after determining that the rotational speed of the motor exceeds a predetermined speed with the motor being driven.

- 24.** The pump of claim 21, wherein the controller is further configured to drive the motor after detecting that the value detected by the capacitive sensor indicates a presence of foreign substance around the pump.
- 25.** The pump of claim 24, wherein detecting that the value detected by the capacitive sensor indicates the presence of the foreign substance around the pump comprises comparing the value detected by the capacitive sensor with a predetermined threshold.
- 26.** The pump of claim 21, wherein the sensor box is provided on a side of the pump cap and above the motor housing.
- 27.** The pump of claim 26, wherein the controller is further configured to drive the motor after detecting that the value detected by the capacitive sensor indicates that the pump is tipped over.
- 28.** The pump of claim 27, wherein at least a portion of the motor housing is filled with motor oil, the motor oil enters the pump cap as the pump tips over, and detecting that the value detected by the capacitive sensor indicates that the pump is tipped over comprises detecting that the value detected by the capacitive sensor indicates that the motor oil is present around the sensor box.
- 29.** The pump of claim 26, wherein the capacitive sensor has a sensing surface, and the sensing surface is secured by a support part to contact an inner side of the sensor box so that the sensing surface faces outward, and the sensor box, the pump cap, and the support part are made of non-conductive material.
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